Arav Palsule

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LinkedIn — GitHub — Portfolio

B.E. Data Science (2027) — CGPA: 8.34/10

Professional Summary

Aspiring Software Engineer with hands-on experience building web applications, backend workflows, REST API integrations, and data-driven software systems. Skilled in Python, JavaScript, TypeScript, SQL, React, and application development. Experienced in developing full-stack applications, integrating databases and external services, implementing application logic, and building automated processing pipelines. Strong foundation in problem-solving, debugging, and software development.

Education

A. P. Shah Institute of Technology, Thane 2023 – 2027
Bachelor of Engineering in Data Science **CGPA: 8.34/10**

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, SQL
Frontend Development: React, Vite, TypeScript
Backend Development: Flask, FastAPI, REST APIs, API Integration, Backend Workflow Development
Databases: MySQL, SQLite, Supabase
Software Development: Application Logic, Data Processing, Debugging, Testing, API Integration
Tools and Platforms: Git, GitHub, Zapier, Zoho
Additional Technologies: LangGraph, LLM Integration, FFmpeg, MoviePy

Projects

Radiator Routes – Full-Stack Travel Planning Platform

- Developed a full-stack travel application with a **five-model workflow** for trip intake, itinerary generation, validation, disruption replanning, and Travel Memory.
- Implemented **Multi-Agent Orchestration** with **LangGraph** and Supabase-backed trip features.
- Added **14 deterministic validation checks** for budget, dates, scheduling, route feasibility, geographic validity, and opening hours.
- Built an LLM-assisted repair flow and **Least Misery** scoring to revise itineraries and balance group preferences.

Technologies Used: React, TypeScript, Vite, Python, LangGraph, Supabase, REST APIs, Groq, WebLLM

CrackX – Road Condition Management System

- Built a role-based reporting workflow for **five user groups**: citizens, Road Safety Officers, contractors, compliance officers, and administrators.
- Integrated **YOLOv8** to classify and localize road cracks and potholes, with severity assessment and duplicate-report checks.
- Implemented contractor assignments, before-and-after photo evidence, and a **two-step approval** flow.
- Connected completed repair records to citizen ratings, written feedback, and time-range activity reports.

Technologies Used: React Native, Expo, TypeScript, Python, FastAPI, YOLOv8, OpenCV, Supabase

CineRec AI – Video Processing and Highlight Generator

- Developed a **five-stage media pipeline** for scene extraction, emotion classification, importance ranking, semantic music matching, and reel composition.
- Implemented asynchronous processing with **FastAPI**, **Celery**, and **Redis**.
- Automated timestamp generation, clip extraction, soundtrack alignment, and final video rendering.

Technologies Used: Python, FastAPI, Celery, Redis, PyTorch, DistilBERT, XGBoost, FFmpeg, MoviePy, Supabase

Dengue Outbreak Prediction – Data Processing Pipeline

- Developed a Python pipeline combining historical dengue cases, weather, geographic attributes, and temporal features for forecasting.
- Implemented preprocessing, feature engineering, and training workflows to compare **four regression algorithms**.
- Evaluated results using chronological holdout and rolling-window validation with **MAE, RMSE, and SMAPE**.

Technologies Used: Python, Pandas, NumPy, Scikit-learn, Random Forest, Extra Trees, HistGradientBoosting, Ridge Regression

Experience

AI Intern – Ennoble Social Innovation
September 2026 – October 2026

- Created custom Claude Skills using structured context engineering to automate business tasks, including PDF formatting according to internal document structures and standards.
- Developed an employee focus-monitoring prototype integrating face detection and emotion analysis to estimate focus scores and identify potential sleep or inactivity events.
- Designed screenshot capture and automated HR reporting workflows to communicate detected events and focus patterns.
- Built business process automations using Zapier, LLMs, and API integrations to streamline repetitive operational tasks.

Achievements

- **Rank 10** – C.R. Rao Statistics & Probability Olympiad: Ranked among 2.5 lakh+ participants across Asia.
- **AIR 3192, AICR 35** – Aryabhatta National Maths Competition 2026: Scored 100% in Round 1 and 80% in Round 2.
- **2nd Runner-Up** – QUASAR 4.0 Hackathon: Delivered a technical solution under a 24-hour development deadline.
- **Elite Rank** – NPTEL Multivariable Calculus: Recognized for mathematical and quantitative problem-solving skills.

Leadership

Founding President – Maths & Applied Science Club — 2026

- Founded and led the club, organizing 25+ technical events to promote mathematics, science, and technical learning among students.

Research Member – DSA Club — 2025 – 2026

- Explored software development, LLMs, RAG, and machine learning pipelines; mentored junior members in technical concepts.